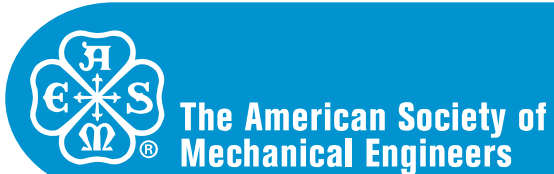


ASME B31.3-2020
(Revision of ASME B31.3-2018)

Process Piping

ASME Code for Pressure Piping, B31

AN INTERNATIONAL PIPING CODE®



ASME B31.3-2020

SUMMARY OF CHANGES

Following approval by the ASME B31 Committee and ASME, and after public review, ASME B31.3-2020 was approved by the American National Standards Institute on September 29, 2020.

ASME B31.3-2020 includes the following changes identified by a margin note, **(20)**.

<i>Page</i>	<i>Location</i>	<i>Change</i>
xx	Introduction	After subpara. (f), paragraph added
2	300.1.4	Added, and subsequent paragraph redesignated
10	Table 300.4	Appendix D deleted
12	301.3.2	Title revised
12	301.3.3	Title revised
12	301.3.4	Title revised
13	302.2.4	Revised
14	302.2.5	Last sentence deleted
16	Table 302.3.3C	General Note revised
19	302.3.6	Subparagraph (a)(1) revised
24	304.3.1	Subparagraphs (a) and (a)(2) revised
31	304.5.2	Subparagraph (b) revised
32	304.7.2	Subparagraphs (b) and (c) revised
34	306.5.2	Revised
34	307.1.2	Last sentence deleted
36	311.2.2	Subparagraph (d) revised
37	314.2.1	Subparagraph (c) revised
40	319.3.4	Subparagraph (b) revised
40	319.3.6	Revised
41	319.4.4	(1) Subparagraphs (a) and (b) revised (2) Subparagraph (c) and eqs. (19) and (20) deleted
43	320.1	Revised
44	320.2	(1) Revised (2) Footnote 9 added
49	Table 323.2.2	Items A-3(b) and B-3 revised
56	Table 323.3.1	(1) Items 4, A-5, and B-5 revised (2) Note (4) deleted
60	Table 326.1	Revised in its entirety
66	328.5.2	Revised in its entirety
66	328.5.4	Subparagraphs (d), (e)(2), and (f) revised
67	Figure 328.5.2A	Title revised
67	Figure 328.5.2B	Title revised
68	Figure 328.5.2C	(1) Title revised (2) Variable for nominal pipe wall thickness revised

<i>Page</i>	<i>Location</i>	<i>Change</i>
69	328.6	Revised
71	330.1	Last paragraph added
73	331.1.1	Subparagraph (a) revised
73	331.1.3	Subparagraph (b)(5)(-a) revised
74	Table 331.1.1	Note (7) revised
73	331.1.6	Subparagraph (c) revised
75	331.2.6	Revised
78	332.4	Last sentence added
90	344.6.2	Subparagraph (a) revised
103	A321.5.1	Subparagraph (a) revised
103	A321.5.2	Revised in its entirety
106	A328.2.1	Subparagraph (b)(6) revised
107	Table A326.1	Standards ASTM D3309, ASTM D2310, ASTM D2447, ASTM D3309, and ASTM F1974 deleted
109	A328.2.5	Nomenclature in subpara. (c)(1) revised
120	M331	Revised
124	K300	Revised in its entirety
124	K300.1	K300.1.4 added, subsequent paragraphs redesignated, and cross-references updated
125	K302.3.1	Subparagraphs (b) and (d) revised
126	K302.3.2	Subparagraphs (b), (c), and (d) revised
127	K302.3.6	Subparagraph (a) revised
128	K304.1.2	Revised
128	K304.1.3	Revised
128	K304.2.4	Revised
129	K304.7.2	Subparagraph (c) revised
130	K304.7.3	Revised
130	K304.8.1	First paragraph and footnote 6 revised
130	K304.8.2	Former subpara. (a) deleted, and subsequent subparagraphs revised and redesignated
130	K304.8.3	Subparagraph (c) revised
130	K304.8.5	First paragraph revised
131	K307	Revised in its entirety
132	K314.2	Subparagraph (b) revised
133	K317.2	Title revised
134	K322.6.3	(1) Subparagraphs (a) and (b) revised (2) Subparagraph (c) deleted
135	K323.2.2	Revised in its entirety
135	K323.3.1	Revised
135	K323.3.4	Revised in its entirety
139	Table K326.1	(1) Standards under “Bolting” and “Metallic Fittings, Valves, and Flanges” revised (2) Note (1)(c) added
140	K328	Revised
140	K328.1	Revised in its entirety

<i>Page</i>	<i>Location</i>	<i>Change</i>
140	K328.2.1	Subparagraphs (b) and (f) revised
140	K328.3.1	Revised in its entirety
141	K330.1	Revised
141	K331	Revised in its entirety
143	K332.4.1	Revised
143	K332.4.2	Subparagraph (a) revised
143	K333	Revised in its entirety
144	K341.3.3	Subparagraphs (a) and (b) revised
144	K341.4.1	(1) Subparagraph (b) revised (2) Subparagraph (c) added
148	K345.1	Subparagraphs (a), (b), and (d) revised
148	K345.2.1	Former footnote 10 deleted
149	K346.1	Revised
149	K346.2	(1) Subparagraph (b) added, and subsequent subparagraphs redesignated (2) Subparagraphs (d) and (f) revised
152	U328	Revised
152	U328.2	Revised in its entirety
152	U328.4.2	Added
154	U341.4.1	(1) First paragraph and subparas. (a) and (b) revised (2) Subparagraph (c) added
159	Specification Index for Appendix A	(1) A381, B166, B462, B649, B688, and B690 revised (2) B547 deleted
163	Notes for Tables A-1, A-1M, A-1A, A-1B, A-2, and A-2M	(1) General Note (f) and in-text table in Note (42) revised (2) Note (44) deleted
167	Table A-1	(1) Under Iron — Castings, for Gray A278 40, Note (9) reference deleted (2) Under Carbon Steel — Pipes and Tubes, for A285 Gr. A A134, A285 Gr. B A134, and A285 Gr. C A134, Type/Grade and UNS No. added (3) For A516 Gr. 60, stress values for 700°F and 800°F revised (4) For A516 Gr. 65, stress values for 850°F and 900°F revised (5) Under Carbon Steel — Pipes (Structural Grade), A283 Gr. A and B deleted (6) A1011 Gr. 30, 33, 36, and 40; A36; A283; and A1011 Gr. 45 and 50 revised (7) Under Carbon Steel — Plates, Bars, Shapes, and Sheets (Structural), A283 A and B deleted (8) A1011 30, 33, 36, 40, 45, and 50; A283 C and D; and A36 revised (9) Under Stainless Steel — Pipes and Tubes, A376 16-8-2H deleted (10) A312 TP310S revised (11) A358 310S deleted (12) For A813, A814, A249, and A312 S31254, Notes and stress values revised

<i>Page</i>	<i>Location</i>	<i>Change</i>
		(13) For A358 S31254, stress values revised
		(14) A789 S32900 deleted
		(15) Under Stainless Steel — Plates and Sheets, A240 439 deleted
		(16) A240 310H revised
		(17) A240 S31254 Notes and stress values revised
		(18) Under Stainless Steel — Forgings and Fittings, A182 F10 deleted
		(19) A182 F310, A403 WP310S, and A403 WP310H revised
		(20) For A182 and A403 S31254, Notes and stress values revised
		(21) A403 S31254, A815 S31803, A403 N08367, and A815 S32750 deleted
		(22) A815 S32101, A815 S32205, and A815 S32760 revised
		(23) A182 S41000 and A182 S41026 deleted
		(24) A182 S32750 revised
		(25) Under Stainless Steel — Bar, A479 S31254 Note and stress values revised
		(26) A479 S32205 and A479 S20910 revised
		(27) Under Stainless Steel — Castings, for A351 CH10, A351 CH20, and A351 CF8C, stress values revised
		(28) Under Nickel and Nickel Alloy — Castings, A494 CX2MW reordered and revised
		(29) A494 CW12MW and A494 CW6M revised
		(30) Under Zirconium and Zirconium Alloy — Pipes and Tubes, B523 R60702, B658 R60702, and B658 R60705 revised
		(31) B523 R60705 deleted
		(32) Under Zirconium and Zirconium Alloy — Plates and Sheets, B551 R60702 and B551 R60705 revised
		(33) Under Zirconium and Zirconium Alloy — Forgings and Bar, B493 R60702, B550 R60702, B493 R60705, and B550 R60705 revised
		(34) Aluminum Alloy — Welded Pipes and Tubes category deleted
242	Table A-1M	(1) Listings for Nominal Composition Fe revised in their entirety
		(2) A134 revised
		(3) For A524, A333, A334, A671, A672, A139, API 5L, and A381, stress values revised
		(4) A516, A515, A1011, A283, A36, and A992 revised
		(5) A376 deleted
		(6) For A312, Type/Grade, UNS No., and Notes revised
		(7) A358 deleted
		(8) For A813, A814, A249, and A312, Notes deleted
		(9) A789 and A240 deleted
		(10) For A240 and A403, Type/Grade, UNS No., and Notes revised
		(11) A182, A403, A815, and A479 revised
		(12) For A351, stress values revised
		(13) Wld. tube B626 added
		(14) A494, B523, B658, B551, B493, and B550 revised

<i>Page</i>	<i>Location</i>	<i>Change</i>
		(15) B547 deleted
349	Table A-1B	B547 deleted
354	Table A-2	(1) A194 4 and A194 4L deleted (2) A193 B8 added
364	Table A-2M	(1) A194 4 and A194 4L deleted (2) A193 B8 added
385	Specification Index for Appendix B	D2447 and D3309 deleted
386	Table B-1	D3309, D2239, D2447, D3035, and F714 deleted
388	Table B-1M	D3309, D2239, D2447, D3035, and F714 deleted
412	Appendix D	Deleted
413	Appendix E	Revised in its entirety
419	F300.1.5	Former para. F300.1.4 editorially redesignated
419	F301.10.2	Revised
420	F302	Added
421	F323	Subparagraph (b) revised
423	F331.1	Revised
424	FA328	Added
428	H301	Revised
428	H302	Revised
429	H303	Revised
431	H304	Revised
437	Appendix J	Cross-references updated
453	Appendix K	(1) ASTM A928 added to Specification Index (2) Page column deleted in Specification Index
455	Notes for Table K-1	(1) General Notes (b) and (d) and Notes (3), (4), and (20) revised (2) General Note (h) added (3) Notes (6) and (12) deleted
456	Table K-1	Revised in its entirety
474	Figure M300	In Col. 1, cross-references to para. K300.1.5 revised to K300.1.6
477	R301	(1) Subparagraph (b) revised (2) Subparagraph (c) deleted
480	Appendix S	Revised in its entirety
495	V304	Revised in its entirety
500	Table W302.1-4	Editorially redesignated
507	Index	Added

Chapter VI

Inspection, Examination, and Testing

340 INSPECTION

340.1 General

This Code distinguishes between examination (see [para. 341](#)) and inspection. Inspection applies to functions performed for the owner by the owner's Inspector or the Inspector's delegates. References in this Code to the "Inspector" are to the owner's Inspector or the Inspector's delegates.

340.2 Responsibility for Inspection

It is the owner's responsibility, exercised through the owner's Inspector, to verify that all required examinations and testing have been completed and to inspect the piping to the extent necessary to be satisfied that it conforms to all applicable examination requirements of the Code and of the engineering design.

340.3 Rights of the Owner's Inspector

The owner's Inspector and the Inspector's delegates shall have access to any place where work concerned with the piping installation is being performed. This includes manufacture, fabrication, heat treatment, assembly, erection, examination, and testing of the piping. They shall have the right to audit any examination, to inspect the piping using any examination method specified by the engineering design, and to review all certifications and records necessary to satisfy the owner's responsibility stated in [para. 340.2](#).

340.4 Qualifications of the Owner's Inspector

(a) The owner's Inspector shall be designated by the owner and shall be the owner, an employee of the owner, an employee of an engineering or scientific organization, or the employee of a recognized insurance or inspection company acting as the owner's agent. The owner's Inspector shall not represent nor be an employee of the piping manufacturer, fabricator, or erector unless the owner is also the manufacturer, fabricator, or erector.

(b) The owner's Inspector shall meet one of the following requirements:

(1) have at least 10 yr of experience in the design, fabrication, or examination of industrial pressure piping. Each 20% of satisfactorily completed work toward an ac-

credited engineering degree shall be considered equivalent to 1 yr of experience, up to 5 yr total.

(2) have a professional engineering registration or nationally recognized equivalent with at least 5 yr of experience in the design, fabrication, or examination of industrial pressure piping.

(3) be a certified welding inspector or a senior certified welding inspector as defined in AWS QC1, Specification for AWS Certification of Welding Inspectors, or nationally recognized equivalent with at least 5 yr of experience in the design, fabrication, or examination of industrial pressure piping.

(4) be an authorized piping inspector as defined in API 570, Piping Inspection Code: In-service Inspection, Rating, Repair, and Alteration of Piping Systems, with at least 5 yr of experience in the design, fabrication, or examination of industrial pressure piping.

(c) In delegating performance of inspection, the owner's Inspector is responsible for determining that a person to whom an inspection function is delegated is qualified to perform that function.

341 EXAMINATION

341.1 General

Examination applies to quality control functions performed by the manufacturer (for components only), fabricator, or erector. Reference in this Code to an examiner is to a person who performs quality control examinations.

341.2 Responsibility for Examination

Inspection does not relieve the manufacturer, the fabricator, or the erector of the responsibility for

(a) providing materials, components, and workmanship in accordance with the requirements of this Code and of the engineering design [see [para. 300\(b\)\(3\)](#)]

(b) performing all required examinations

(c) preparing suitable records of examinations and tests for the Inspector's use

341.3 Examination Requirements

341.3.1 General. Prior to initial operation, each piping installation, including components and workmanship, shall be examined in accordance with the applicable

requirements of [para. 341](#). The type and extent of any additional examination required by the engineering design, and the acceptance criteria to be applied, shall be specified. Joints not included in examinations required by [para. 341.4](#) or by the engineering design are accepted if they pass the leak test required by [para. 345](#).

(a) For P-Nos. 3, 4, 5A, 5B, 5C, and 15E materials, examinations shall be performed after completion of heat treatment. However, examinations need not be repeated on welds or portions of welds that are subjected to additional heat treatments and have not been repaired by welding.

(b) For a welded branch connection, the examination of, and any necessary repairs to, the pressure-containing weld shall be completed before any reinforcing pad or saddle is added.

341.3.2 Acceptance Criteria. Acceptance criteria shall be as stated in the engineering design and shall at least meet the applicable requirements stated below.

(a) *Welds.* See [Figure 341.3.2](#) for typical weld imperfections.

(1) For radiography and visual, see [Table 341.3.2](#).

(2) For magnetic particle, see [para. 344.3.2](#).

(3) For liquid penetrant, see [para. 344.4.2](#).

(4) For ultrasonic, see [para. 344.6.2](#).

(b) *Castings.* Acceptance criteria for castings are specified in [para. 302.3.3](#).

341.3.3 Defective Components and Workmanship. Defects (imperfections of a type or magnitude not acceptable by the criteria specified in [para. 341.3.2](#)) shall be repaired, or the defective item or work shall be replaced. Discontinuities detected outside the area required to be examined during weld joint examinations should be evaluated and resolved in a manner acceptable to the owner and designer.

Examination shall be as follows:

(a) When the defective item or work is repaired, the repaired portion of the item or work shall be examined. The examination shall use the same methods and acceptance criteria employed for the original examination. See also [para. 341.3.1\(a\)](#).

(b) When the defective item or work is replaced, the new item or work used to replace the defective item or work shall be examined. The examination shall use any method and applicable acceptance criteria that meet the requirements for the original examination. See also [para. 341.3.1\(a\)](#).

341.3.4 Progressive Sampling for Examination. When required spot or random examination reveals a defect, then

(a) two additional samples of the same kind (if welded or bonded joints, by the same welder, bonder, or operator) from the original designated lot shall be given the same type of examination

(b) if the items examined as required by (a) above are acceptable, the defective item shall be repaired or replaced and reexamined as specified in [para. 341.3.3](#), and all items represented by these two additional samples shall be accepted, but

(c) if any of the items examined as required by (a) above reveals a defect, two further samples of the same kind shall be examined for each defective item found by that sampling

(d) if all the items examined as required by (c) above are acceptable, the defective item(s) shall be repaired or replaced and reexamined as specified in [para. 341.3.3](#), and all items represented by the additional sampling shall be accepted, but

(e) if any of the items examined as required by (c) above reveals a defect, all items represented by the progressive sampling shall be either

(1) repaired or replaced and reexamined as required, or

(2) fully examined and repaired or replaced as necessary, and reexamined as necessary to meet the requirements of this Code

(f) If any of the defective items are repaired or replaced, reexamined, and a defect is again detected in the repaired or replaced item, continued progressive sampling in accordance with (a), (c), and (e) is not required based on the new defects found. The defective item(s) shall be repaired or replaced and reexamined until acceptance as specified in [para. 341.3.3](#). Spot or random examination (whichever is applicable) is then performed on the remaining unexamined joints.

341.4 Extent of Required Examination

341.4.1 Examination — Normal Fluid Service. Piping in Normal Fluid Service shall be examined to the extent specified herein or to any greater extent specified in the engineering design. Acceptance criteria are as stated in [para. 341.3.2](#) and in [Table 341.3.2](#), for Normal Fluid Service unless otherwise specified.

(a) *Visual Examination.* At least the following shall be examined in accordance with [para. 344.2](#):

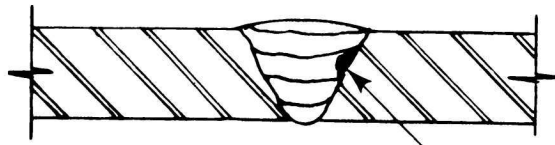
(1) sufficient materials and components, selected at random, to satisfy the examiner that they conform to specifications and are free from defects.

(2) at least 5% of fabrication, as defined in [para. 300.2](#).

(3) 100% of all completed welds, except those in components made in accordance with a listed standard. See [para. 341.5.1\(a\)](#) for examination of longitudinal welds required to have a joint factor, E_j , of 0.90.

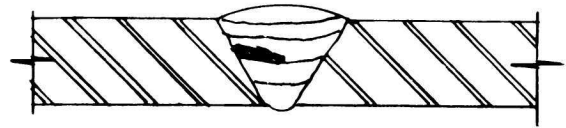
(4) random examination of the assembly of threaded, bolted, and other joints to satisfy the examiner that they conform to the applicable requirements of [para. 335](#). When pneumatic testing is to be performed, all threaded, bolted, and other mechanical joints shall be examined.

Figure 341.3.2 Typical Weld Imperfections

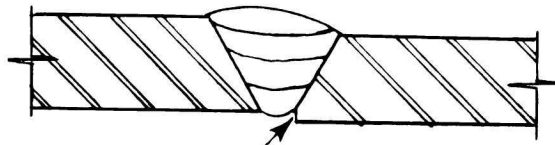


Lack of fusion between weld bead and base metal

(a) Side Wall Lack of Fusion

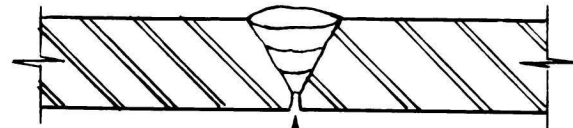


(b) Lack of Fusion Between Adjacent Passes



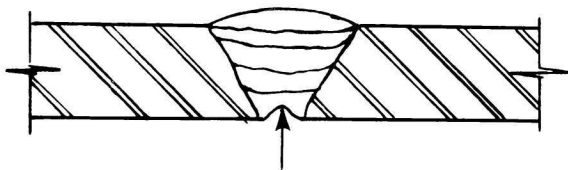
Incomplete filling at root on one side only

(c) Incomplete Penetration due to Internal Misalignment



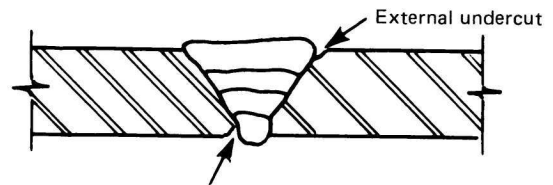
Incomplete filling at root

(d) Incomplete Penetration of Weld Groove



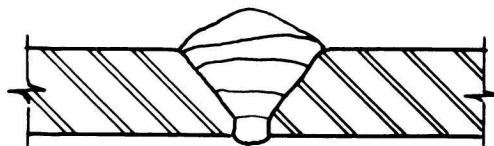
Root bead fused to both inside surfaces but center of root slightly below inside surface of pipe (not incomplete penetration)

(e) Concave Root Surface (Suck-Up)



Internal undercut

(f) Undercut



(g) Excess External Reinforcement

Table 341.3.2 Acceptance Criteria for Welds — Visual and Radiographic Examination

Criteria (A to M) for Types of Welds and for Service Conditions [Note (1)]													Examination Methods										
Normal and Category M Fluid Service			Severe Cyclic Conditions				Category D Fluid Service																
Girth, Miter Groove, and Branch Connection Welds [Note (2)]	Longitudinal Groove Weld [Note (3)]	Fillet Weld [Note (4)]	Girth, Miter Groove, and Branch Connection Welds [Note (2)]	Longitudinal Groove Weld [Note (3)]	Fillet Weld [Note (4)]	Girth and Miter Groove Welds	Longitudinal Groove Weld [Note (3)]	Fillet Weld [Note (4)]	Branch Connection Weld [Note (2)]	Weld Imperfection	Visual	Radiography											
											✓	✓											
											A	A	A	A	A	A	A	A	A	A	Crack	✓	✓
											A	A	A	A	A	A	C	A	N/A	A	Lack of fusion	✓	✓
											B	A	N/A	A	A	N/A	C	A	N/A	B	Incomplete penetration	✓	✓
											E	E	N/A	D	D	N/A	N/A	N/A	N/A	N/A	Rounded Indications	...	✓
											G	G	N/A	F	F	N/A	N/A	N/A	N/A	N/A	Linear indications	...	✓
											H	A	H	A	A	A	I	A	H	H	Undercutting	✓	✓
											A	A	A	A	A	A	A	A	A	A	Surface porosity or exposed slag inclusion [Note (5)]	✓	...
											N/A	N/A	N/A	J	J	J	N/A	N/A	N/A	N/A	Surface finish	✓	...
K	K	N/A	K	K	N/A	K	K	N/A	K	Concave surface, concave root, or burn-through	✓	✓											
L	L	L	L	L	L	M	M	M	M	Weld reinforcement or internal protrusion	✓	...											

GENERAL NOTES:

- Weld imperfections are evaluated by one or more of the types of examination methods given, as specified in [paras. 341.4.1, 341.4.2, 341.4.3, and M341.4](#), or by the engineering design.
- "N/A" indicates the Code does not establish acceptance criteria or does not require evaluation of this kind of imperfection for this type of weld.
- Check (✓) indicates examination method generally used for evaluating this kind of weld imperfection.
- Ellipsis (...) indicates examination method not generally used for evaluating this kind of weld imperfection.

NOTES:

- Criteria given are for required examination. More-stringent criteria may be specified in the engineering design. See also [paras. 341.5 and 341.5.3](#).
- Branch connection weld includes pressure containing welds in branches and fabricated laps.
- Longitudinal groove weld includes straight and spiral (helical) seam. Criteria are not intended to apply to welds made in accordance with a standard listed in [Table A-1, Table A-1M, or Table 326.1](#). Alternative Leak Test requires examination of these welds; see [Para. 345.9](#).
- Fillet weld includes socket and seal welds, and attachment welds for slip-on flanges, branch reinforcement, and supports.
- These imperfections are evaluated only for welds ≤5 mm (³/₁₆ in.) in nominal thickness.

Criterion Value Notes for Table 341.3.2

Criterion		Acceptable Value Limits [Note (1)]	
Symbol	Measure		
A	Extent of imperfection	Zero (no evident imperfection)	
B	Cumulative length of incomplete penetration	≤38 mm (1.5 in.) in any 150 mm (6 in.) weld length or 25% of total weld length, whichever is less	
C	Cumulative length of lack of fusion and incomplete penetration	≤38 mm (1.5 in.) in any 150 mm (6 in.) weld length or 25% of total weld length, whichever is less	
D	Size and distribution of rounded indications	See ASME BPVC, Section VIII, Division 1, Appendix 4 [Note (2)]	
E	Size and distribution of rounded indications	For $\bar{T}_w \leq 6$ mm ($\frac{1}{4}$ in.), limit is same as D [Note (2)] For $\bar{T}_w > 6$ mm ($\frac{1}{4}$ in.), limit is $1.5 \times D$ [Note (2)]	
F	Linear indications	$\leq \bar{T}_w/3$	
G	Individual length	≤ 2.5 mm ($\frac{3}{32}$ in.) and $\leq \bar{T}_w/3$	
	Individual width	$\leq \bar{T}_w$ in any 12 $\bar{T}_w$ weld length [Note (2)]	
	Cumulative length		
	Linear indications	$\leq 2\bar{T}_w$	
H	Individual length	≤ 3 mm ($\frac{1}{8}$ in.) and $\leq \bar{T}_w/2$	
	Individual width	$\leq 4\bar{T}_w$ in any 150 mm (6 in.) weld length [Note (2)]	
	Cumulative length	≤ 1 mm ($\frac{1}{32}$ in.) and $\leq \bar{T}_w/4$	
	Depth of undercut	≤ 38 mm (1.5 in.) in any 150 mm (6 in.) weld length or 25% of total weld length, whichever is less	
I	Cumulative length of internal and external undercut	≤ 1.5 mm ($\frac{1}{16}$ in.) and $\leq [\bar{T}_w/4 \text{ or } 1 \text{ mm } (\frac{1}{32} \text{ in.})]$	
J	Depth of undercut	≤ 38 mm (1.5 in.) in any 150 mm (6 in.) weld length or 25% of total weld length, whichever is less	
K	Surface roughness	≤ 12.5 μm (500 $\mu\text{in.}$) R_a in accordance with ASME B46.1	
L	Depth of surface concavity, root concavity, or burn-through	Total joint thickness, including weld reinforcement, $\geq \bar{T}_w$ [Notes (3) and (4)]	
M	Height of reinforcement or internal protrusion [Note (5)] in any plane through the weld shall be within limits of the applicable height value in the tabulation at right, except as provided in Note (6). Weld metal shall merge smoothly into the component surfaces.	For $\bar{T}_w$, mm (in.)	
		Height, mm (in.)	
		≤ 6 ($\frac{1}{4}$)	
		≤ 1.5 ($\frac{1}{16}$)	
M	Height of reinforcement or internal protrusion [Note (5)] as described in L. Note (6) does not apply.	> 6 ($\frac{1}{4}$), ≤ 13 ($\frac{1}{2}$)	
		≤ 3 ($\frac{1}{8}$)	
		> 13 ($\frac{1}{2}$), ≤ 25 (1)	
		≤ 4 ($\frac{5}{32}$)	
M		> 25 (1)	
		≤ 5 ($\frac{3}{16}$)	
		Limit is twice the value applicable for L above	

NOTES:

(1) Where two limiting values are separated by “and,” the lesser of the values determines acceptance. Where two sets of values are separated by “or,” the larger value is acceptable. $\bar{T}_w$ is the nominal wall thickness of the thinner of two components joined by a butt weld.

Criterion Value Notes for Table 341.3.2 (Cont'd)

NOTES: (Cont'd)

- (2) Porosity and inclusions such as slag or tungsten are defined as rounded indications where the maximum length is three times the width or less. These indications may be circular, elliptical, or irregular in shape; may have tails; and may vary in density. Indications where the length is greater than three times the width are defined as linear indications and may also be slag, porosity, or tungsten.
 - (3) For circumferential groove welded joints in pipe, tube, and headers made entirely without the addition of filler metal, external concavity shall not exceed the lesser of 1 mm ($1/32$ in.) or 10% of the joint nominal thickness. The contour of the concavity shall blend smoothly with the base metal. The total joint thickness, including any reinforcement, shall not be less than the minimum wall thickness, t_m .
 - (4) For radiography, acceptability may be determined by comparing the density of the image through the affected area to the density through the adjacent base metal ($\bar{T}_w$). If digital radiography is used, brightness comparison may be utilized. A density or brightness darker than the adjacent base metal is cause for rejection.
 - (5) For groove welds, height is the lesser of the measurements made from the surfaces of the adjacent components; both reinforcement and internal protrusion are permitted in a weld. For fillet welds, height is measured from the theoretical throat. [Figure 328.5.2A](#); internal protrusion does not apply.
 - (6) For welds in aluminum alloy only, internal protrusion shall not exceed the following values:
 - (a) 1.5 mm ($1/16$ in.) for thickness ≤ 2 mm ($5/64$ in.)
 - (b) 2.5 mm ($3/32$ in.) for thickness > 2 mm and ≤ 6 mm ($1/4$ in.)
- For external reinforcement and for greater thicknesses, see the tabulation for symbol L.

(5) random examination during erection of piping, including checking of alignment, supports, and cold spring.

(6) examination of erected piping for evidence of defects that would require repair or replacement, and for other evident deviations from the intent of the design.

(b) Other Examination

(1) Not less than 5% of circumferential butt and miter groove welds shall be examined fully by random radiography in accordance with [para. 344.5](#) or by random ultrasonic examination in accordance with [para. 344.6](#). The welds to be examined in each designated lot shall include the work product of each welder or welding operator whose welds are part of the lot. The work of welders depositing only tack welds need not be represented as part of the lot. Welds shall also be selected to maximize coverage of intersections with longitudinal joints. When a circumferential weld with an intersecting longitudinal weld(s) is examined, at least the adjacent 38 mm (1½ in.) of each intersecting weld shall be examined. In-process examination in accordance with [para. 344.7](#) may be substituted for all or part of the radiographic or ultrasonic examination on a weld-for-weld basis if specified in the engineering design or specifically authorized by the Inspector.

(2) Not less than 5% of all brazed joints shall be examined by in-process examination in accordance with [para. 344.7](#), the joints to be examined being selected to ensure that the work of each brazer making the production joints is included.

(c) Certifications and Records. The examiner shall be assured, by examination of certifications, records, and other evidence, that the materials and components are of the specified grades and that they have received required heat treatment, examination, and testing. The examiner shall provide the Inspector with a certification that all the quality control requirements of the Code and of the engineering design have been carried out.

341.4.2 Examination — Category D Fluid Service.

Piping and piping elements for Category D Fluid Service as designated in the engineering design shall be visually examined in accordance with [para. 344.2](#) to the extent necessary to satisfy the examiner that components, materials, and workmanship conform to the requirements of this Code and the engineering design. Acceptance criteria are as stated in [para. 341.3.2](#) and in [Table 341.3.2](#), for Category D fluid service, unless otherwise specified.

341.4.3 Examination — Severe Cyclic Conditions.

Piping to be used under severe cyclic conditions shall be examined to the extent specified herein or to any greater extent specified in the engineering design. Acceptance criteria are as stated in [para. 341.3.2](#) and in [Table 341.3.2](#), for severe cyclic conditions, unless otherwise specified.

(a) Visual Examination. The requirements of [para. 341.4.1\(a\)](#) apply with the following exceptions:

(1) All fabrication shall be examined.

(2) All threaded, bolted, and other joints shall be examined.

(3) All piping erection shall be examined to verify dimensions and alignment. Supports, guides, and points of cold spring shall be checked to ensure that movement of the piping under all conditions of startup, operation, and shutdown will be accommodated without undue binding or unanticipated constraint.

(b) Other Examination. All circumferential butt and miter groove welds and all fabricated branch connection welds comparable to those shown in [Figure 328.5.4E](#) shall be examined by 100% radiography in accordance with [para. 344.5](#), or (if specified in the engineering design) by 100% ultrasonic examination in accordance with [para. 344.6](#). Socket welds and branch connection welds that are not radiographed shall be examined by magnetic particle or liquid penetrant methods in accordance with [para. 344.3](#) or [344.4](#).

(c) In-process examination in accordance with [para. 344.7](#), supplemented by appropriate nondestructive examination, may be substituted for the examination required in [\(b\)](#) above on a weld-for-weld basis if specified in the engineering design or specifically authorized by the Inspector.

(d) Certification and Records. The requirements of [para. 341.4.1\(c\)](#) apply.

341.4.4 Examination — Elevated Temperature Fluid Service. Piping in Elevated Temperature Fluid Service shall be examined to the extent specified herein or to any greater extent specified in the engineering design. Acceptance criteria are as stated in [para. 341.3.2](#) and in [Table 341.3.2](#), for Normal Fluid Service, unless the requirements for severe cyclic conditions apply or otherwise specified.

(a) Visual Examination. The requirements of [para. 341.4.1\(a\)](#) apply with the following exceptions:

(1) All fabrication shall be examined.

(2) All threaded, bolted, and other joints shall be examined.

(3) All piping erection shall be examined to verify dimensions and alignment. Supports, guides, and points of cold spring shall be checked to ensure that movement of the piping under all conditions of startup, operation, and shutdown will be accommodated without undue binding or unanticipated constraint.

(b) Additional Examination. The examination requirements of [para. 341.4.1\(b\)](#) apply with the following exceptions:

(1) Fabrication for longitudinal and spiral (helical seam) welds in P-No. 4 and P-No. 5 materials, except those in components made in accordance with a listed specification, shall be examined by 100% radiography

in accordance with [para. 344.5](#), or by 100% ultrasonic examination in accordance with [para. 344.6](#).

(2) Socket welds and branch connection welds in P-No. 4 and P-No. 5 materials that are not radiographed or ultrasonically examined shall be examined by magnetic particle or liquid penetrant methods in accordance with [para. 344.3](#) or [344.4](#).

(c) *Additional Examination Required for Autogenous Welds, Without Filler Metal, in Austenitic Stainless Steel and Austenitic High Nickel Alloys.* Autogenously welded pipe shall receive nondestructive examination in accordance with the material specification. Autogenously welded expansion joint bellows shall be examined in accordance with [para. X302.2.2\(c\)](#).

(d) *Certification and Records.* The requirements of [para. 341.4.1\(c\)](#) apply.

341.5 Supplementary Examination

Any of the methods of examination described in [para. 344](#) may be specified by the engineering design to supplement the examination required by [para. 341.4](#). The extent of supplementary examination to be performed and any acceptance criteria that differ from those in [para. 341.3.2](#) shall be specified in the engineering design.

341.5.1 Spot Radiography

(a) *Longitudinal Welds.* Spot radiography for longitudinal groove welds required to have a weld joint factor E_j of 0.90 requires examination by radiography in accordance with [para. 344.5](#) of at least 300 mm (1 ft) in each 30 m (100 ft) of weld for each welder or welding operator. Acceptance criteria are those stated in [Table 341.3.2](#) for radiography under Normal Fluid Service.

(b) *Circumferential Butt Welds and Other Welds.* It is recommended that the extent of examination be not less than one shot on one in each 20 welds for each welder or welding operator. Unless otherwise specified, acceptance criteria are as stated in [Table 341.3.2](#) for radiography under Normal Fluid Service for the type of joint examined.

(c) *Progressive Sampling for Examination.* The provisions of [para. 341.3.4](#) are applicable.

(d) *Welds to Be Examined.* The locations of welds and the points at which they are to be examined by spot radiography shall be selected or approved by the Inspector.

341.5.2 Hardness Tests. Hardness tests are not required to verify proper heat treatment except as otherwise specified in the engineering design.

341.5.3 Examinations to Resolve Uncertainty. Any method may be used to resolve doubtful indications. Acceptance criteria shall be those for the required examination.

342 EXAMINATION PERSONNEL

342.1 Personnel Qualification and Certification

Personnel performing nondestructive examination to the requirements of this Code shall be qualified and certified for the method to be utilized following a procedure as described in ASME BPVC, Section V, Article 1, T-120(e) or (f).

342.2 Specific Requirement

For in-process examination, the examinations shall be performed by personnel other than those performing the production work.

343 EXAMINATION PROCEDURES

Any examination shall be performed in accordance with a written procedure that conforms to one of the methods specified in [para. 344](#), including special methods (see [para. 344.1.2](#)). Procedures shall be written as required in ASME BPVC, Section V, Article 1, T-150. The employer shall make the examination procedures employed available to the Inspector.

344 TYPES OF EXAMINATION

344.1 General

344.1.1 Methods. Except as provided in [para. 344.1.2](#), any examination required by this Code, by the engineering design, or by the Inspector shall be performed in accordance with one of the methods specified herein.

344.1.2 Special Methods. If a method not specified herein is to be used, it and its acceptance criteria shall be specified in the engineering design in enough detail to permit qualification of the necessary procedures and examiners.

344.1.3 Definitions. The following terms apply to any type of examination:

*100% examination*¹: complete examination of all of a specified kind of item in a designated lot of piping¹

*random examination*²: complete examination of a percentage of a specified kind of item in a designated lot of piping¹

¹ A designated lot is that quantity of piping to be considered in applying the requirements for examination in this Code. The quantity or extent of a designated lot should be established by agreement between the contracting parties before the start of work. More than one kind of designated lot may be established for different kinds of piping work. See Pipe Fabrication Institute Standard ES-48, Random Examination, for examples of lot selection.

² Random or spot examination will not ensure a fabrication product of a prescribed quality level throughout. Items not examined in a lot of piping represented by such examination may contain defects that further examination could disclose. Specifically, if all radiographically disclosable weld defects must be eliminated from a lot of piping, 100% radiographic examination must be specified.

*spot examination*²: a specified partial examination of each of a specified kind of item in a designated lot of piping,¹ e.g., of part of the length of all shop-fabricated welds in a lot of jacketed piping

*random spot examination*²: a specified partial examination of a percentage of a specified kind of item in a designated lot of piping¹

344.2 Visual Examination

344.2.1 Definition. Visual examination is the direct observation of the external and internal portions of components, joints, and other piping elements that are readily accessible or can be exposed to view before, during, or after manufacture, fabrication, assembly, erection, examination, or testing. This examination includes verification of Code and engineering design requirements for materials, components, dimensions, joint preparation, alignment, welding, bonding, brazing, bolting, threading, or other joining method, supports, assembly, and erection.

344.2.2 Method. Visual examination shall be performed in accordance with ASME BPVC, Section V, Article 9. Examination shall be by the direct visual technique. The use of a remote visual technique and the acceptance criteria beyond the requirements of this Code shall be a matter of agreement between contracting parties prior to the start of fabrication.

Records of individual visual examinations are not required except for those of in-process examinations as specified in [para. 344.7](#).

344.3 Magnetic Particle Examination

344.3.1 Method. Examination of castings is covered in [para. 302.3.3](#). Magnetic particle examination of welds and of components other than castings shall be performed in accordance with ASME BPVC, Section V, Article 7.

344.3.2 Acceptance Criteria. Magnetic particle indications are caused by the attraction of the test media to surface or near-surface discontinuities in the area under test. However, all such indications are not necessarily imperfections, since excessive roughness, magnetic permeability variations, etc., may produce nonrelevant indications. Inadvertent accumulation of particles not related to magnetic attraction is classified as a false indication. Indications shall be verified as being relevant, nonrelevant, or false. Additional surface preparation and/or other test methods may be used as needed to verify the relevance of an indication.

An indication of an imperfection may be larger than the imperfection that causes it; however, the size of the indication is the basis for acceptance evaluation. Only indications that have any dimension greater than 1.5 mm ($\frac{1}{16}$ in.) shall be considered relevant.

(a) *Indications*

(1) A linear indication is one having a length greater than three times its width.

(2) A rounded indication is one of circular or elliptical shape with a length equal to or less than three times its width.

(b) *Examination.* All surfaces to be examined shall be free of

(1) relevant linear indications

(2) relevant rounded indications >5.0 mm ($\frac{3}{16}$ in.)

(3) four or more relevant rounded indications in a line separated by 1.5 mm ($\frac{1}{16}$ in.) or less, edge to edge

344.4 Liquid Penetrant Examination

344.4.1 Method. Examination of castings is covered in [para. 302.3.3](#). Liquid penetrant examination of welds and of components other than castings shall be performed in accordance with ASME BPVC, Section V, Article 6.

344.4.2 Acceptance Criteria. Liquid penetrant indications are caused by the bleed-out of a visible or fluorescent dye from a surface discontinuity in the area under test. However, all such indications are not necessarily imperfections, since excessive roughness, poor surface preparation, etc., may produce nonrelevant indications. Inadvertent evidence of penetrant not related to actual bleed-out is classified as a false indication. Indications shall be verified as being relevant, nonrelevant, or false. Additional surface preparation and/or other test methods may be used as needed to verify the relevance of an indication.

An indication of an imperfection may be larger than the imperfection that causes it; however, the size of the indication is the basis for acceptance evaluation. Only indications that have any dimension greater than 1.5 mm ($\frac{1}{16}$ in.) shall be considered relevant.

(a) *Indications*

(1) A linear indication is one having a length greater than three times its width.

(2) A rounded indication is one of circular or elliptical shape with a length equal to or less than three times its width.

(b) *Examination.* All surfaces to be examined shall be free of

(1) relevant linear indications

(2) relevant rounded indications >5.0 mm ($\frac{3}{16}$ in.)

(3) four or more relevant rounded indications in a line separated by 1.5 mm ($\frac{1}{16}$ in.) or less, edge to edge

344.5 Radiographic Examination

344.5.1 Method. Radiography of castings is covered in [para. 302.3.3](#). Radiography of welds and of components other than castings shall be performed in accordance with ASME BPVC, Section V, Article 2. For the purpose of image quality indicator (IQI) selection, for welds with reinforcement, the thickness used shall be the nominal wall

thickness, $\bar{T}_w$, plus the allowable external reinforcement and internal reinforcement (protrusion) combined.

344.5.2 Extent of Radiography

(a) *100% Radiography.* This applies only to girth and miter groove welds and to fabricated branch connection welds comparable to Figure 328.5.4E, unless otherwise specified in the engineering design.

(b) *Random Radiography.* This applies only to girth and miter groove welds.

(c) *Spot Radiography.* This requires a single exposure radiograph in accordance with para. 344.5.1 at a point within a specified extent of welding. For girth, miter, and branch groove welds, the minimum requirement is

(1) for sizes $\leq$ DN 65 (NPS $2\frac{1}{2}$), a single elliptical exposure encompassing the entire weld circumference

(2) for sizes $>$ DN 65, the lesser of 25% of the inside circumference or 152 mm (6 in.)

For longitudinal welds, the minimum requirement is 152 mm (6 in.) of weld length.

344.6 Ultrasonic Examination

344.6.1 Method. Examination of castings is covered in para. 302.3.3; other product forms are not covered. Ultrasonic examination of welds shall be performed in accordance with ASME BPVC, Section V, Article 4, except that the alternative specified in (a) and (b) below is permitted for basic calibration blocks specified in T-434.2.1 and T-434.3.

(a) When the basic calibration blocks have not received heat treatment in accordance with T-434.1.5, transfer methods shall be used to correlate the responses from the basic calibration block and the component. Transfer is accomplished by noting the difference between responses received from the same reference reflector in the basic calibration block and in the component and correcting for the difference.

(b) The reference reflector may be a V-notch (which must subsequently be removed), an angle beam search unit acting as a reflector, or any other reflector that will aid in accomplishing the transfer.

(c) When the transfer method is chosen as an alternative, it shall be used, at the minimum

(1) for sizes $\leq$ DN 50 (NPS 2), once in each ten welded joints examined

(2) for sizes $>$ DN 50 and $\leq$ DN 450 (NPS 18), once in each 1.5 m (5 ft) of welding examined

(3) for sizes $>$ DN 450, once for each welded joint examined

(d) Each type of material and each size and wall thickness shall be considered separately in applying the transfer method. In addition, the transfer method shall be used at least twice on each type of weld joint.

(e) The reference level for monitoring discontinuities shall be modified to reflect the transfer correction when the transfer method is used.

344.6.2 Acceptance Criteria. Acceptance criteria shall be as described in (a) or (b). (20)

(a) A discontinuity is unacceptable if the amplitude of the indication exceeds the reference level and its length exceeds

(1) 6 mm ($\frac{1}{4}$ in.) for $\bar{T}_w \leq 19$ mm ($\frac{3}{4}$ in.)

(2) $\bar{T}_w/3$ for 19 mm ($\frac{3}{4}$ in.) $< \bar{T}_w \leq 57$ mm ($2\frac{1}{4}$ in.)

(3) 19 mm ($\frac{3}{4}$ in.) for $\bar{T}_w > 57$ mm ($2\frac{1}{4}$ in.)

(b) The fracture-mechanics-based ultrasonic examination acceptance criteria in Appendix R may be used if all requirements of Appendix R are met.

344.7 In-Process Examination

344.7.1 Definition. In-process examination comprises examination of the following, as applicable:

(a) joint preparation and cleanliness

(b) preheating

(c) fit-up, joint clearance, and internal alignment prior to joining

(d) variables specified by the joining procedure, including filler material

(1) (for welding) position and electrode

(2) (for brazing) position, flux, brazing temperature, proper wetting, and capillary action

(e) (for welding) condition of the root pass after cleaning — external and, where accessible, internal — aided by liquid penetrant or magnetic particle examination when specified in the engineering design

(f) (for welding) slag removal and weld condition between passes

(g) appearance of the finished joint

344.7.2 Method. The examination is visual, in accordance with para. 344.2, unless additional methods are specified in the engineering design.

345 TESTING

345.1 Required Leak Test

Prior to initial operation, and after completion of the applicable examinations required by para. 341, each piping system shall be tested to ensure tightness. The test shall be a hydrostatic leak test in accordance with para. 345.4 except as provided herein.

(a) At the owner's option, a piping system in Category D fluid service may be subjected to an initial service leak test in accordance with para. 345.7, in lieu of the hydrostatic leak test.

(b) Where the owner considers a hydrostatic leak test impracticable, either a pneumatic test in accordance with para. 345.5 or a combined hydrostatic-pneumatic test in

accordance with [para. 345.6](#) may be substituted, recognizing the hazard of energy stored in compressed gas.

(c) Where the owner considers both hydrostatic and pneumatic leak testing impracticable, the alternative specified in [para. 345.9](#) may be used if both of the following conditions apply:

- (1) a hydrostatic test would
 - (-a) damage linings or internal insulation
 - (-b) contaminate a process that would be hazardous, corrosive, or inoperative in the presence of moisture
 - (-c) require significant support modifications for the hydrostatic test load or
 - (-d) present the danger of brittle fracture due to low metal temperature during the test
- (2) a pneumatic test would
 - (-a) present an undue hazard of possible release of energy stored in the system or
 - (-b) present the danger of brittle fracture due to low metal temperature during the test
- (d) Unless specified in the engineering design, lines open to the atmosphere, such as vents or drains downstream of the last shutoff valve, need not be leak tested.

345.2 General Requirements for Leak Tests

Requirements in [para. 345.2](#) apply to more than one type of leak test.

345.2.1 Limitations on Pressure

(a) *Reduced Test Pressure.* If the test pressure would produce a circumferential or longitudinal stress (based on minimum pipe wall thickness) in excess of yield strength at test temperature or is greater than 1.5 times the component rating at test temperature, the test pressure may be reduced to the maximum pressure that will not exceed the lesser of the yield strength or 1.5 times a component rating at test temperature. [See [para. 302.3.2.](#)] For metallic bellows expansion joints, see [Appendix X, para. X302.2.3\(a\)](#).

(b) *Test Fluid Expansion.* If a pressure test is to be maintained for a period of time and the test fluid in the system is subject to thermal expansion, precautions shall be taken to avoid excessive pressure.

(c) *Preliminary Pneumatic Test.* A preliminary test using air at no more than 170 kPa (25 psi) gage pressure may be made prior to hydrostatic testing to locate major leaks.

345.2.2 Other Test Requirements

(a) *Examination for Leaks.* The leak test pressure shall be maintained for at least 10 min and then all joints and connections shall be examined for leaks. The test pressure may be reduced to not less than the design pressure while performing this examination.

(b) *Heat Treatment.* Leak tests shall be conducted after any heat treatment has been completed.

(c) *Low Test Temperature.* The possibility of brittle fracture shall be considered when conducting leak tests at metal temperatures near the ductile–brittle transition temperature.

345.2.3 Special Provisions for Testing

(a) *Piping Components and Subassemblies.* Piping components and subassemblies may be tested either separately or as assembled piping.

(b) *Flanged Joints.* Flanged joints used to connect piping components and subassemblies that have previously been tested, and flanged joints at which a blank or blind is used to isolate equipment or other piping during a test, need not be leak tested in accordance with [para. 345.1](#).

(c) *Closure Welds.* The final weld connecting piping systems or components that have been successfully tested in accordance with [para. 345](#) need not be leak tested provided the weld is examined in-process in accordance with [para. 344.7](#) and passes with 100% radiographic examination in accordance with [para. 344.5](#) or 100% ultrasonic examination in accordance with [para. 344.6](#).

(d) *Instrument Connections.* Threaded joints, tubing joints, or a combination of these joints used to connect instruments to previously leak tested piping need not be leak tested in accordance with [para. 345.1](#).

(e) See also [Appendix F, para. F345.2.3](#).

345.2.4 Externally Pressured Piping

(a) Except as provided in (b) below, piping systems subject to external pressure shall be tested at an internal gage pressure 1.5 times the external differential pressure, but not less than 105 kPa (15 psi).

(b) As an alternative to leak testing under internal pressure, piping systems designed for vacuum service only may be subjected to a vacuum leak test method, technique, and acceptance criteria specified by the owner. The vacuum leak test shall be performed following a written procedure complying with the applicable technical requirements of ASME BPVC, Section V, Article 10. Leak-testing personnel shall be qualified and certified as required by ASME BPVC, Section V, Article 1, T-120(e) or (f).

345.2.5 Jacketed Piping

(a) The internal line shall be leak tested on the basis of the internal or external design pressure, whichever results in a higher test pressure. This test must be performed before the jacket is completed if it is necessary to provide visual access to joints of the internal line as required by [para. 345.3.1](#).

(b) The jacket shall be leak tested in accordance with [para. 345.1](#) based on the jacket design conditions. The test pressure is permitted to be lower when so specified in the engineering design.

345.2.6 Repairs or Additions After Leak Testing. If repairs or additions are made following the leak test, the affected piping shall be retested, except that for minor repairs or additions the owner may waive retest requirements when precautionary measures are taken to assure sound construction.

345.2.7 Test Records. Records shall be made of each piping system during the testing, including

- (a) date of test
- (b) identification of piping system tested
- (c) test fluid
- (d) test pressure
- (e) certification of results by examiner

These records need not be retained after completion of the test if a certification by the Inspector that the piping has satisfactorily passed pressure testing as required by this Code is retained.

345.3 Preparation for Leak Test

345.3.1 Joints Exposed

(a) Except as provided in (b) and (c) below, all joints, welds (including structural attachment welds to pressure-containing components), and bonds shall be left uninsulated and exposed for examination during leak testing.

(b) Joints previously tested in accordance with this Code may be insulated or covered.

(c) At the owner's option, joints in Category D Fluid Service that are subject to a hydrostatic leak test (para. 345.4) or an initial service leak test (para. 345.7) may be insulated and have protective weather sheathing installed prior to leak testing. Consideration shall be given to increasing the test period to allow time for possible leakage to pass through the insulation and weather sheathing.

(d) All joints may be primed and painted prior to leak testing unless a sensitive leak test (para. 345.8) is required.

345.3.2 Temporary Supports. Piping designed for vapor or gas shall be provided with additional temporary supports, if necessary, to support the weight of test liquid.

345.3.3 Piping With Expansion Joints

(a) Unrestrained expansion joints depend on external main anchors to resist pressure thrust forces. Except as limited in (c), a piping system containing unrestrained expansion joints shall be leak tested without any temporary restraints in accordance with para. 345 up to 150% of the expansion joint design pressure. If the required test pressure exceeds 150% of the expansion joint design pressure and the main anchors are not designed to resist the pressure thrust forces at the required test pressure, for that portion of the test when the pressure exceeds 150% of the expansion joint design pressure, the expansion joint shall either

be temporarily removed or temporary restraints shall be added to resist the pressure thrust forces.

(b) Self-restrained metallic bellows expansion joints (i.e., tied, hinged, pressure balanced, etc.) have restraint hardware designed to resist the pressure thrust forces. Except as limited in (c), a piping system containing self-restrained expansion joints shall be leak tested in accordance with para. 345. A self-restrained expansion joint previously shop tested by the manufacturer in accordance with Appendix X may be excluded from the system to be leak tested, except when a sensitive leak test in accordance with para. 345.8 is required. Restraint hardware for all types of expansion joints shall be designed for the pressure thrust forces at the test pressure.

(c) When a metallic bellows expansion joint is installed in the piping system subject to a leak test and the leak test pressure determined in accordance with para. 345 exceeds the pressure of the test performed by the manufacturer in accordance with Appendix X, the required leak test pressure shall be reduced to the manufacturer's test pressure.

345.3.4 Limits of Tested Piping. Equipment that is not to be tested shall be either disconnected from the piping or isolated by blinds or other means during the test. A valve may be used provided the valve (including its closure mechanism) is suitable for the test pressure.

345.4 Hydrostatic Leak Test

345.4.1 Test Fluid. The fluid shall be water unless there is the possibility of damage due to freezing or to adverse effects of water on the piping or the process (see para. F345.4.1). In that case another suitable nontoxic liquid may be used. If the liquid is flammable, its flash point shall be at least 49°C (120°F), and consideration shall be given to the test environment.

345.4.2 Test Pressure. Except as provided in para. 345.4.3, the hydrostatic test pressure at every point in a metallic piping system shall be as follows:

- (a) not less than 1.5 times the design pressure.
- (b) when the design temperature is greater than the test temperature, the minimum test pressure, at the point under consideration, shall be calculated using eq. (24).

$$P_T = 1.5 P S_T / S \quad (24)$$

where

P = internal design gage pressure

P_T = minimum test gage pressure

S = allowable stress at component design temperature for the prevalent pipe material; see Table A-1 or Table A-1M

S_T = allowable stress at test temperature for the prevalent pipe material; see Table A-1 or Table A-1M

(c) in those cases where the piping system may not include pipe itself, any other component in the piping system, other than pipe-supporting elements and bolting, may be used to determine the S_T/S ratio based on the applicable allowable stresses obtained from Table A-1 or Table A-1M. In those cases where the piping system may be made up of equivalent lengths of more than one material, the S_T/S ratio shall be based on the minimum calculated ratio of the included materials.

345.4.3 Hydrostatic Test of Piping With Vessels as a System³

(a) Where the test pressure of piping attached to a vessel is the same as or less than the test pressure for the vessel, the piping may be tested with the vessel at the piping test pressure.

(b) Where the test pressure of the piping exceeds the vessel test pressure, and it is not considered practicable to isolate the piping from the vessel, the piping and the vessel may be tested together at the vessel test pressure, provided the owner approves and the vessel test pressure is not less than 77% of the piping test pressure calculated in accordance with para. 345.4.2(b).

345.5 Pneumatic Leak Test

345.5.1 Precautions. Pneumatic testing involves the hazard of released energy stored in compressed gas. Particular care must therefore be taken to minimize the chance of brittle failure during a pneumatic leak test. Test temperature is important in this regard and must be considered when the designer chooses the material of construction. See para. 345.2.2(c) and Appendix F, paras. F323.4 and F345.5.1.

345.5.2 Pressure Relief Device. A pressure relief device shall be provided, having a set pressure not higher than the test pressure plus the lesser of 345 kPa (50 psi) or 10% of the test pressure.

345.5.3 Test Fluid. The gas used as test fluid, if not air, shall be nonflammable and nontoxic.

345.5.4 Test Pressure. The test pressure shall be not less than 1.1 times the design pressure and shall not exceed the lesser of

(a) 1.33 times the design pressure

(b) the pressure that would exceed 90% of the pressure described in para. 345.2.1(a)

345.5.5 Procedure. The pressure shall be gradually increased until a gage pressure that is the lesser of one-half the test pressure or 170 kPa (25 psi) is attained, at which time a preliminary check shall be made, including examination of joints in accordance with para. 341.4.1(a).

Thereafter, the pressure shall be gradually increased in steps until the test pressure is reached, holding the pressure at each step long enough to equalize piping strains. The pressure shall then be reduced to the design pressure before examining for leakage in accordance with para. 345.2.2(a).

345.6 Hydrostatic-Pneumatic Leak Test

If a combination hydrostatic-pneumatic leak test is used, the requirements of para. 345.5 shall be met, and the pressure in the liquid filled part of the piping shall not exceed the limits stated in para. 345.4.2.

345.7 Initial Service Leak Test

This test is applicable only to piping in Category D Fluid Service, at the owner's option. See para. 345.1(a).

345.7.1 Test Fluid. The test fluid is the service fluid.

345.7.2 Procedure. During or prior to initial operation, the pressure shall be gradually increased in steps until the operating pressure is reached, holding the pressure at each step long enough to equalize piping strains. A preliminary check shall be made as described in para. 345.5.5 if the service fluid is a gas or vapor.

345.7.3 Examination for Leaks. The examination for leaks required by para. 345.2.2(a) shall be conducted while the system is at operating pressure. It is permissible to omit examination for leaks of joints and connections previously tested in accordance with this Code.

345.8 Sensitive Leak Test

345.8.1 Precautions. The precautions described in para. 345.5.1 shall be considered when applicable.

345.8.2 Method. The test shall be the Bubble Test — Direct Pressure Technique in accordance with ASME BPVC, Section V, Article 10, Mandatory Appendix I or another leak test method that has a demonstrated sensitivity not less than 10^{-3} std mL/s under test conditions.

When the Bubble Test — Direct Pressure Technique is used

(a) the test pressure shall be at least the lesser of 105 kPa (15 psi) or 25% of the design pressure.

(b) the pressure shall be gradually increased until a gage pressure equal to the lesser of one-half the test pressure or 170 kPa (25 psi) is attained, at which time a preliminary check shall be made. Then the pressure shall be gradually increased in steps until the test pressure is reached, the pressure being held long enough at each step to equalize piping strains.

345.9 Alternative Leak Test

The following procedures and leak test method may be used only under the conditions stated in para. 345.1(c).

³ The provisions of para. 345.4.3 do not affect the pressure test requirements of any applicable vessel code.

345.9.1 Examination of Welds. Welds, including those used in the manufacture of welded pipe and fittings, that have not been subjected to hydrostatic or pneumatic leak tests in accordance with this Code, shall be examined as follows:

(a) Circumferential, longitudinal, and spiral (helical seam) groove welds shall be 100% radiographed in accordance with [para. 344.5](#) or 100% ultrasonically examined in accordance with [para. 344.6](#).

(b) All welds, including structural attachment welds, not covered in (a) above, shall be examined using the liquid penetrant method ([para. 344.4](#)) or, for magnetic materials, the magnetic particle method ([para. 344.3](#)).

345.9.2 Flexibility Analysis. A flexibility analysis of the piping system shall have been made in accordance with the requirements of [para. 319.4.2\(b\)](#), if applicable, or [paras. 319.4.2\(c\)](#) and [319.4.2\(d\)](#).

345.9.3 Test Method. The system shall be subjected to a sensitive leak test in accordance with [para. 345.8](#).

346 RECORDS

346.2 Responsibility

It is the responsibility of the piping designer, the manufacturer, the fabricator, and the erector, as applicable, to prepare the records required by this Code and by the engineering design.

346.3 Retention of Records

Unless otherwise specified by the engineering design, the following records shall be retained for at least 5 yr after the record is generated for the project:

- (a) examination procedures
- (b) examination personnel qualifications
- (c) examination reports